Confidence-Shielded, Evaluator-Conditioned Fusion for Risk-Controlled Infrared–Visible Object Detection

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Abstract

Infrared and visible fusion is often treated as a universally beneficial preprocessing step, yet one fused image can improve one detector and degrade another. We introduce Confidence-Shielded, Evaluator-Conditioned Fusion (CS-ECF), a label-free policy that scores visible, consensus-detail fusion (CDF), PIAFusion, and SPGFusion candidates with the target detector. Each candidate is scored by the mean of up to five highest confidences. CS-ECF uses a fused candidate only when its score reaches 1.5 times that of visible; otherwise, it retains visible. This margin was fixed before confirmatory testing.

We evaluated CS-ECF on a fixed-split subset of LLVIP, a grouped six-class split of M3FD, and an official-test subset of DroneVehicle, using 12 confirmatory checkpoints. Nine of the 12 checkpoint contrasts were positive, but none survived Holm correction and the exact sign test was not significant ($p = 0.073$). The dataset-balanced AP50 gain was 0.0133, with a 95% training-run-by-group interval of [0.0026, 0.0261]. Against Always CDF, the balanced difference was +0.0102, but its interval $[-0.00057, 0.02980]$ included zero, so superiority was not established. Removing the shield increased the magnitude of the worst AP50 loss from 0.0040 to 0.0584. Two zero-shot evaluators improved AP50 over visible by 0.021–0.025, but neither separated from Always CDF. An LLVIP-fine-tuned YOLOv9t lost 0.0108 AP50 relative to visible; shielding nevertheless improved the no-shield selector by 0.0153. CS-ECF did not transfer uniformly across evaluators. The evidence supports bounded, evaluator-dependent risk control, not uniform improvement, CDF dominance, or low-latency deployment. Candidate generation and four detector passes took approximately 0.71 s per image pair.

Keywords: infrared–visible image fusion, object detection, evaluator-conditioned fusion, confidence calibration, risk control, thermal imaging

1. Introduction

Visible and infrared imaging provide complementary evidence. Visible cameras capture fine spatial detail and colour, whereas infrared sensors can retain target contrast under poor illumination or smoke. These properties have motivated pixel-, feature-, and network-based fusion methods for surveillance, autonomous perception, and thermal imaging systems [1–10]. For object detection, however, the value of a fused image depends on more than visual appeal. It is determined by the change in detector outputs under a specified operating condition.

Current studies commonly select one fused representation and evaluate it with one detector on one benchmark. This design assumes that a globally preferred fusion result remains preferable across detector architectures and training seeds. The assumption is fragile because detectors differ

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in receptive-field size, confidence calibration, training data, and sensitivity to high-frequency detail [11, 12]. A fusion rule that improves one evaluator may therefore reduce the average precision (AP) of another on the same image.

Per-image selection offers a natural response: evaluate several fusion candidates and choose the one with the strongest detector response. Unconstrained selection, however, can turn small calibration differences into frequent input switches and severe AP losses when a spurious high-confidence response is mistaken for better detection. The task is therefore to decide when the evidence is strong enough to depart from the visible-image fallback, which is related to the classical reject option in recognition [13].

Recent infrared–visible fusion and detection studies have optimized shared fusion networks, cross-modal feature fusion, or hierarchical detectors [14–17]. General-purpose detectors provide the evaluator side of this problem [18, 19]. These studies show why fusion and detection should be considered jointly, but they generally train or optimize a detector–fusion system. They do not provide a label-free arbitration rule with a threshold fixed independently of confirmatory testing and evaluated at the checkpoint, image, and scene or frame-group levels.

We formulate infrared–visible fusion selection as evaluator-conditioned input arbitration. CS-ECF evaluates four candidates with the target detector, computes a top-five confidence score, and accepts a fused candidate only when its score reaches 1.5 times that of the visible input. The threshold is selected on validation data and frozen before testing. We evaluate the policy on LLVIP, M3FD, and DroneVehicle using checkpoint-, image-, and group-aware uncertainty. We also compare it with fixed fusion and a no-shield selector, and we report its cost and failure boundaries. The contributions are:

1. a label-free, evaluator-conditioned formulation of fusion-input selection;
2. a confidence shield that reduces catastrophic switching while preserving positive dataset-balanced gains;
3. a three-dataset risk-boundary evaluation that separates image, group, and checkpoint uncertainty and reports the people-class concentration, the absence of a separated advantage over Always CDF, and a fine-tuned-evaluator counterexample to universal portability.

2. Related work

2.1. Fusion representations

Classical infrared–visible fusion methods combine complementary evidence using handcrafted transforms or representations. Representative examples include multi-scale top-hat transforms, non-subsampled contourlet transforms with compressed sensing, two-scale saliency decomposition, and three-scale framelet or shear representations [3, 20–22]. Sparse-representation methods instead describe local image content with learned or overcomplete dictionaries before applying a coefficient-level fusion rule [23, 24]. These methods explicitly preserve target contrast, edges, and local saliency. Their fusion rules are typically designed for image-level visual quality rather than for the ranking assigned by a particular detector.

Learned fusion models replace handcrafted combination rules with trainable feature extraction and reconstruction. DenseFuse, IFCNN, U2Fusion, and RFN-Nest are representative encoder–decoder or residual designs [4, 6–8]. Adversarial formulations add discriminative losses for texture and target preservation [5, 25–27]. Wavelet-based dual-path networks and transformer-based designs have since added multi-scale context and long-range interaction [28–31], while diffusion models introduced generative priors for fusion [9, 10]. These methods broaden the candidate design space but generally produce a single fused output. They do not specify how to select among several available candidates for a fixed deployed detector.

2.2. Task-aware fusion and detection

Task-aware fusion introduces an illumination, semantic, or saliency signal into the transfer rule. PIAFusion conditions fusion on illumination [32]; SPGFusion and MTFusion incorporate semantic priors [33, 34]; and classification-saliency or masked autoencoder designs use auxiliary representations to decide which features to retain [35, 36]. Such priors make the transfer more selective, but a proxy such as illumination or segmentation does not guarantee a better confidence ordering for every detector.

One-stage and DETR-style detectors exhibit different confidence behaviour and receptive-field structure [18, 19, 37, 38]. Recent infrared–visible systems have integrated fusion with detection through shared backbones, cross-modal feature exchange, saliency masking, or hierarchical fusion [14–17, 39, 40]. Joint optimisation aligns the fused representation with the detector, but the learned coupling is specific to a training objective. It does not itself provide a label-free rule for choosing among already generated fusion candidates when the target checkpoint or data composition changes.

2.3. Evaluation and selection boundary

Fusion quality has commonly been assessed with reference or no-reference image metrics [1, 2]. Those scores summarise source retention or perceptual quality; they do not identify which of several inputs will produce the higher AP for a particular detector. Multispectral detection studies show that thermal evidence can improve robustness under poor illumination [41, 42], but gains depend on architecture, training data, and the operating regime [11, 12]. The same dependence appears visible across the detector checkpoints examined here.

The decision to depart from a visible-image fallback resembles the classical reject option [13], but the object of the decision is an input candidate rather than a class prediction. The threshold must therefore be specified independently of test-set performance, fixed before confirmation, and evaluated against both average gain and worst-case loss. A fused-image score or a single benchmark pairing alone leaves that decision boundary untested.

2.4. Positioning

CS-ECF does not introduce another image-fusion operator. It treats an existing fusion library as a set of alternative inputs and uses the target detector as a label-free evaluator. Unlike task-aware networks, it requires neither a new fusion model nor joint detector–fusion training. Unlike fixed image-quality rankings, it measures detector utility directly while accounting for checkpoints, scene or frame groups, and independent evaluators. The contribution is therefore an arbitration policy and a risk-boundary evaluation, not a claim that any single fused representation is universally superior.

3. Methods

3.1. Problem formulation

We denote a registered visible–infrared pair by x_v and x_i, and the target detector checkpoint by $\mathcal{D}$. A candidate $c \in \mathcal{C}$ produces an input image x_c, where $\mathcal{C} = \{\mathrm{vis}, \mathrm{cdf}, \mathrm{pia}, \mathrm{spg}\}$. The policy returns one candidate per pair without access to ground-truth boxes:

$$g(x_v, x_i; \mathcal{D}) \in \mathcal{C}. \tag{1}$$

The final detection output is $\mathcal{D}(x_{g(x_v,x_i;\mathcal{D})})$. Because the same detector is used to evaluate candidates and to produce final detections, the policy requires no separate label oracle.

3.2. Fusion candidates

The visible input is always retained as a safety candidate. CDF combines a pixel-mean image and an IFCNN image using sign-consistent high-frequency details. We denote the pixel-mean and IFCNN images by x_m and x_f, respectively, and let G_σ denote Gaussian blur with $\sigma = 5$. The corresponding detail images are $d_m = x_m - G_\sigma * x_m$ and $d_i = x_f - G_\sigma * x_f$. Their consensus detail is

$$d(x) = \begin{cases} \operatorname{sign}(d_m + d_i)\min(|d_m|, |d_i|), & d_m d_i > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

A thermal-validity mask M combines the infrared top-hat response, infrared intensity, visible luminance, connected components, and spatial smoothing. The quantile $q_{\mathrm{TH}} = 0.95$ marks the top-hat response above which a pixel is considered thermally salient. This quantity differs from the candidate score q_c defined below. We applied a 31×31 elliptical top-hat kernel and an infrared floor of 120. Candidate regions also required a visible-luminance cutoff at the 75th percentile, 5×5 morphological opening, and connected components of at least 64 pixels. The surviving mask was then dilated and smoothed with 31×31 kernels. Shared detail was injected into the LAB luminance channel as

$$L' = L + \alpha M d. \tag{3}$$

The frozen settings are $q_{\mathrm{TH}} = 0.95$, $\alpha = 0.20$, and $\sigma = 5$. PIAFusion and SPGFusion were generated using their official model definitions and pretrained weights under the same image-pair protocol [4, 32, 33].

3.3. Confidence-shielded arbitration

For candidate c, let $s_{c,1} \geq s_{c,2} \geq \cdots$ denote detector confidences sorted in descending order. We define the candidate score as

$$q_c = \frac{1}{\min(K, n_c)} \sum_{j=1}^{\min(K,n_c)} s_{c,j}, \qquad K = 5, \tag{4}$$

where n_c is the number of detections. Candidates with no detections receive $q_c = -1$. The highest-scoring candidate is $c^* = \arg\max_c q_c$, with ties broken in favour of visible. The selected input is

$$c = \begin{cases} c^*, & c^* \neq \text{vis and } q_{c^*} \geq 1.5 q_{\mathrm{vis}}, \\ \text{vis}, & \text{otherwise.} \end{cases} \tag{5}$$

The complete inference path, including visible fallback, is shown in Fig. 1. We fixed the four-candidate library and the top-five score before confirmatory testing. The top-five mean reduces reliance on a single detection while excluding the low-confidence tail. The candidate set includes the visible fallback and representative region-selected, learned, and published fusion methods. Before testing, we evaluated seven margins from 1.0 to 2.0 $\{1.00, 1.10, 1.20, 1.30, 1.50, 1.75, 2.00\}$ on 150 LLVIP validation images and 400 M3FD validation images, using two detectors per dataset. Our objective was to maximize the dataset-balanced validation gain while ensuring that no evaluator lost more than 0.0005 AP50. Under the fixed top-five rule, 1.5 was the first margin to satisfy this constraint and produced the largest balanced validation gain. At this margin, 4.7% of validation decisions switched to a fused input, compared with 11.4% at 1.2. Table 1 reports a post-freeze sensitivity analysis over other score depths and margins. The expanded grid was descriptive and was not used to reselect the policy after test results were observed. The complete pre-freeze search appears in the Supplementary Material and is summarised in Fig. 2.

The validation gains are selection diagnostics from four evaluator cells. They are not estimates of the 12-checkpoint confirmatory effect and should not be compared directly with the test-set point estimate. The margin was not tuned on DroneVehicle, and its public validation partition was not used for policy selection. DroneVehicle therefore serves as an external domain-transfer check.

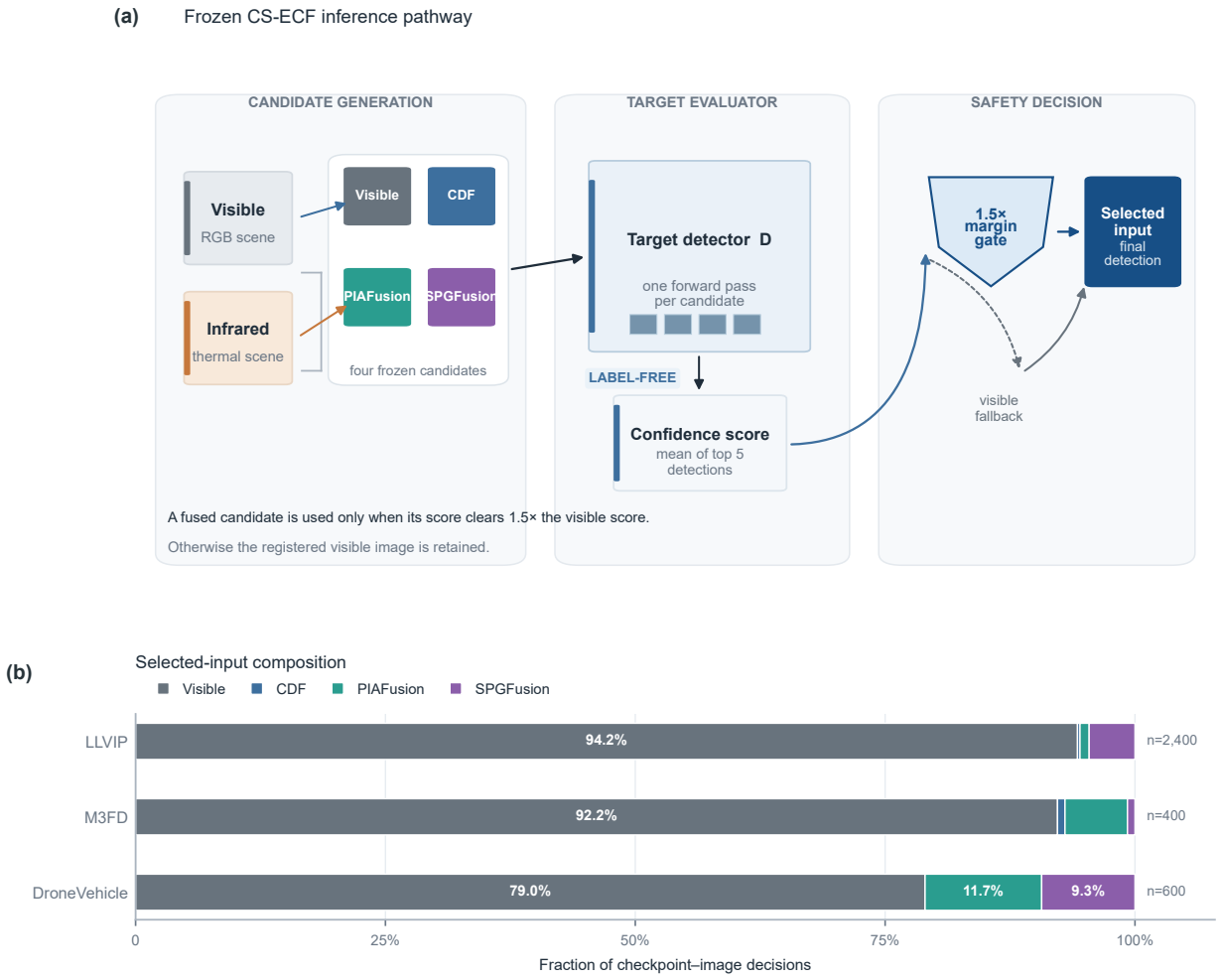


Figure 1: **Confidence-shielded, evaluator-conditioned fusion.** (**a**), The registered visible–infrared pair is used to generate visible, CDF, PIAFusion, and SPGFusion inputs. The target detector computes each candidate score as the mean of up to five highest confidences. A fused candidate is selected only when its score is at least 1.5 times the visible score; otherwise, the visible input is retained. No ground-truth labels are used for arbitration. (**b**), Composition of the frozen input decisions across all checkpoint–image trials for LLVIP (8×300), M3FD (2×200), and the DroneVehicle official test subset (2×300). Source data are provided in `Figure1_source.csv`.

Table 1: Validation sensitivity to score depth and visible margin. Each entry reports the dataset-balanced AP50 gain, with the worst-evaluator gain in parentheses, across the four LLVIP and M3FD validation evaluators. The frozen policy is $K = 5$ with margin 1.5. The expanded grid is descriptive and was not used to reselect the policy after confirmatory testing.

Margin	$K = 1$	$K = 5$	$K = 10$
1.2	-0.0014 (-0.0048)	-0.0002 (-0.0056)	-0.0029 (-0.0073)
1.5	-0.0000 (-0.0004)	**+0.0008 (-0.0001)**	+0.0004 (-0.0010)
2.0	-0.0000 (-0.0004)	+0.0001 (-0.0004)	+0.0016 (+0.0000)
2.5	0.0000 (0.0000)	+0.0001 (0.0000)	+0.0000 (-0.0004)

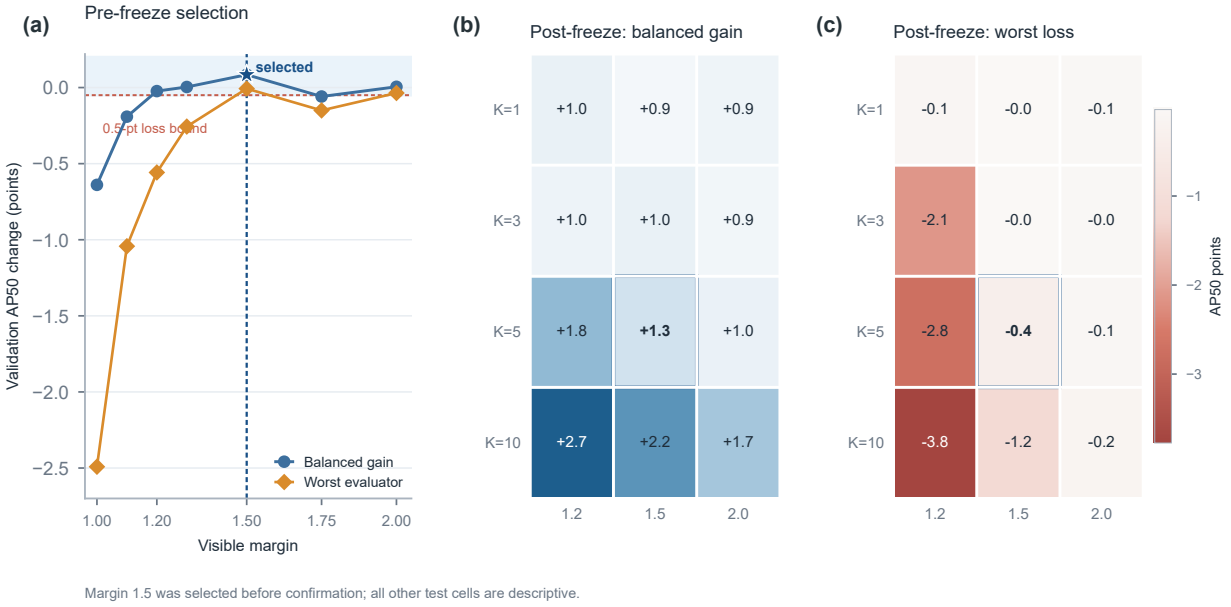


Figure 2: **Threshold selection and post-freeze sensitivity.** (**a**), Pre-freeze validation search over seven visible margins at $K = 5$. The selected policy maximizes the dataset-balanced validation gain subject to no evaluator losing more than 0.0005 AP50. (**b**, **c**), Post-freeze descriptive grids show how dataset-balanced gain and the worst checkpoint loss varied with score depth K and margin. Margin 1.5 was fixed before confirmatory testing; the expanded grid was not used to reselect the policy. Source data are provided in `Figure2_source.csv`.

3.4. Datasets and evaluators

The official LLVIP partitions are larger than the subsets used here [43]. We uniformly shuffled the official test pairs with non-empty labels using seed 0 and used the first 300 as the test subset. The release contains 3,463 test pairs with non-empty labels, so 3,163 eligible pairs were not evaluated. From the official training partition, we selected the lexicographically first 1,500 pairs with non-empty labels as an experiment pool. We assigned the first 1,350 pairs to model training and the remaining 150 pairs to validation. No image stem occurs across train, validation, and test partitions. The three partitions contain 4,479, 419, and 744 pedestrian boxes, respectively.

The eight LLVIP confirmatory checkpoints comprised visible-trained YOLO11n, a modality-agnostic YOLO11n control, YOLO26n seeds 0–2, RT-DETR-l, and YOLOv8n seeds 0 and 1. We used YOLO11n seed 0 for each YOLO11n family and retained four independently trained runs per family for seed sensitivity analysis. All best checkpoints were selected on the 150 validation images, not on the test subset.

M3FD uses a grouped six-class split containing 1,600 training, 400 validation, and 200 test images [44]. We assigned frames to consecutive 100-frame groups with a fixed group seed, preventing adjacent frames from crossing splits. The released manifest specifies the two test groups. We selected one YOLO26n and one YOLOv8n visible-trained six-class evaluator on the validation groups.

We evaluated 300 DroneVehicle image pairs from three fixed consecutive 100-frame groups

Table 2: Training, validation, and confirmatory test partitions. DroneVehicle training and validation counts were not reconstructed locally; its checkpoints use the public partitions specified by the released model recipe. All evaluated DroneVehicle images come from the official test archive.

Dataset	Train	Validation	Test	Evaluation groups
LLVIP	1,350	150	300	7 scene groups
M3FD	1,600	400	200	2 frame groups
DroneVehicle	public train	public validation	300	3 frame groups

Table 3: Evaluator inventory. The "Checkpoints" column includes only the confirmatory models used in the policy comparison. Additional independent runs are retained for sensitivity analysis.

Dataset	Evaluator	Training input	Seed / checkpoint	Checkpoints
LLVIP	YOLO11n	visible	seed 0	1
LLVIP	YOLO11n	visible + infrared	seed 0	1
LLVIP	YOLO26n	visible	seeds 0, 1, 2	3
LLVIP	RT-DETR-l	visible	seed 0	1
LLVIP	YOLOv8n	visible	seeds 0, 1	2
M3FD	YOLO26n	visible, six classes	seed 0	1
M3FD	YOLOv8n	visible, six classes	seed 0	1
Drone-Vehicle	YOLO11xOBB	VIS, vehicle OBB	best/last, one run	2

of the official test archive [45]. The groups were selected with a predefined seed of 2037 before evaluation. The YOLO11x-OBB best and last checkpoints come from one 150-epoch training run (batch size 32, 640-pixel input, seed 0). Because these checkpoints share training data, we treat them as one training-run cluster rather than independent architectures. The source archive has 4,430,343,899 bytes and SHA-256 `94BF47E493E7A57FD5D900439F614F339E7B6C7181E4D781BCFEBAEB963FFD8E` to identify the exact release used for the test groups.

External evaluator stress tests. We also applied the frozen selector to three checkpoints that were not used to construct the policy or select its margin. YOLOv9t and YOLOv10s were COCO-pretrained models without LLVIP fine-tuning, and only their person-class detections were retained [37, 38]. A third YOLOv9t was initialized from the same COCO weights and fine-tuned for a prespecified 40 epochs on the 1,350 LLVIP visible training images. Training used AdamW, a learning rate of 0.002 with three-epoch warm-up and cosine decay, weight decay 5×10^{-4}, batch size 16, 640-pixel inputs, horizontal flipping, and brightness/contrast jitter. We used a reproducible PIL-based loader and the Ultralytics detection loss. We used the checkpoint from epoch 40 as the final model and did not use test results for epoch selection.

Detection evaluation. Predictions use a 640-pixel inference size and a confidence floor of 0.001. For YOLO models, non-maximum suppression uses an IoU threshold of 0.7. At each class and each IoU threshold from 0.50 to 0.95 in steps of 0.05, predicted boxes are assigned to ground-truth boxes by one-to-one maximum-cardinality matching over valid pairs. The resulting true-positive flags and confidence scores are pooled across images within a checkpoint. We integrate the resulting precision-recall curve over 101 points using Ultralytics 8.4.189 [46]. AP50 is averaged over the classes present in the dataset, and AP50–95 additionally averages the ten IoU thresholds.

3.5. Statistical analysis

Point estimates use pooled AP50 after concatenating detections across test images within a checkpoint. Image-bootstrap intervals resample images within a Bootstrap intervals for scenes or

consecutive frame groups are the primary clustered intervals. A detector checkpoint is the unit for architecture-level summaries. Images and groups are repeated observations within a checkpoint and are never treated as independent training runs. Checkpoints from one training run remain one training-run cluster. The dataset-balanced estimand averages within each dataset and then weights LLVIP, M3FD, and DroneVehicle equally. Because M3FD contains only two test groups and LLVIP contains seven scenes, the corresponding group intervals are finite-cluster summaries.

The confirmatory multiplicity family contains the 12 prespecified CS-ECF-minus-visible checkpoint contrasts. For each checkpoint, we used 2,000 paired scene- or frame-group bootstrap resamples to compute two-sided bootstrap p-values. We then applied Holm correction at family-wise $\alpha = 0.05$ and report Benjamini–Hochberg q values. We also summarise the number of positive checkpoints with an exact sign test, which we do not treat as a substitute for effect sizes. The dataset-balanced estimate uses 10,000 training-run-cluster-by-group resamples, and 20 independent 5,000-draw reruns were used to check interval stability. The dataset-balanced contrast between CS-ECF and Always CDF is a prespecified secondary estimand. Its per-dataset values are descriptive rather than a separate confirmatory family. The three external-evaluator contrasts form an exploratory stress-test family, and Benjamini–Hochberg correction is applied across these checks.

We define detection-level ECE using ten fixed confidence bins and also report the Brier score for the external evaluators. These quantities summarise calibration for matched detections; they are not calibrated class probabilities. Candidate selection uses the top-five confidence mean without calibration adjustment.

The training-run-by-group bootstrap used the parameters above with `numpy.random.default_rng(2219)` in Python 3.10.22 and NumPy 2.2.6. Spearman association tests used SciPy 1.15.3. The Supplementary Material provides the complete split counts, all per-checkpoint CS-ECF and Always CDF results, the validation-policy record, and the multiplicity, baseline, class, calibration, and thermal analyses. The public analysis script reconstructs each checkpoint and rejects it if its AP does not agree with the published aggregate table before computing the paired interval.

3.6. Implementation and cost

Using 384×512 fusion inputs, FP32 precision, and a batch size of one on an NVIDIA RTX 2080 Ti, the CDF pipeline required 22.3 ms, PIAFusion 37.7 ms, and SPGFusion 516.3 ms. Four serial YOLO11x-OBB forward passes at 640-pixel input required 130.0 ms, giving an estimated serial total of 706.3 ms. This estimate excludes model loading or swapping, data loading, and final detection post-processing. The additional peak activation memory of SPGFusion was 1453.5 MiB; per-stage memory values should not be summed as an end-to-end peak.

4. Results

4.1. Fusion utility is evaluator-dependent

Across the 12 confirmatory checkpoints, the same candidate set produced both positive and negative AP50 changes (Fig. 3). Nine of the 12 CS-ECF checkpoints had positive point estimates, and the dataset-balanced AP50 gain was 0.0133 with a training-run-by-group 95% interval of [0.0026, 0.0261]. The exact sign test for 9/12 positive contrasts gave $p = 0.073$. After Holm correction across the 12-checkpoint family, no individual contrast remained significant at 0.05. The smallest raw p-value, for LLVIP/YOLO11n, was 0.005 (Holm-adjusted $p = 0.060$). We therefore interpret the balanced estimate as a small average effect and do not claim checkpoint-specific significance (Fig. 4). The no-shield selector had a larger dataset-balanced mean (0.0332), but its worst AP50 change was −0.0584.

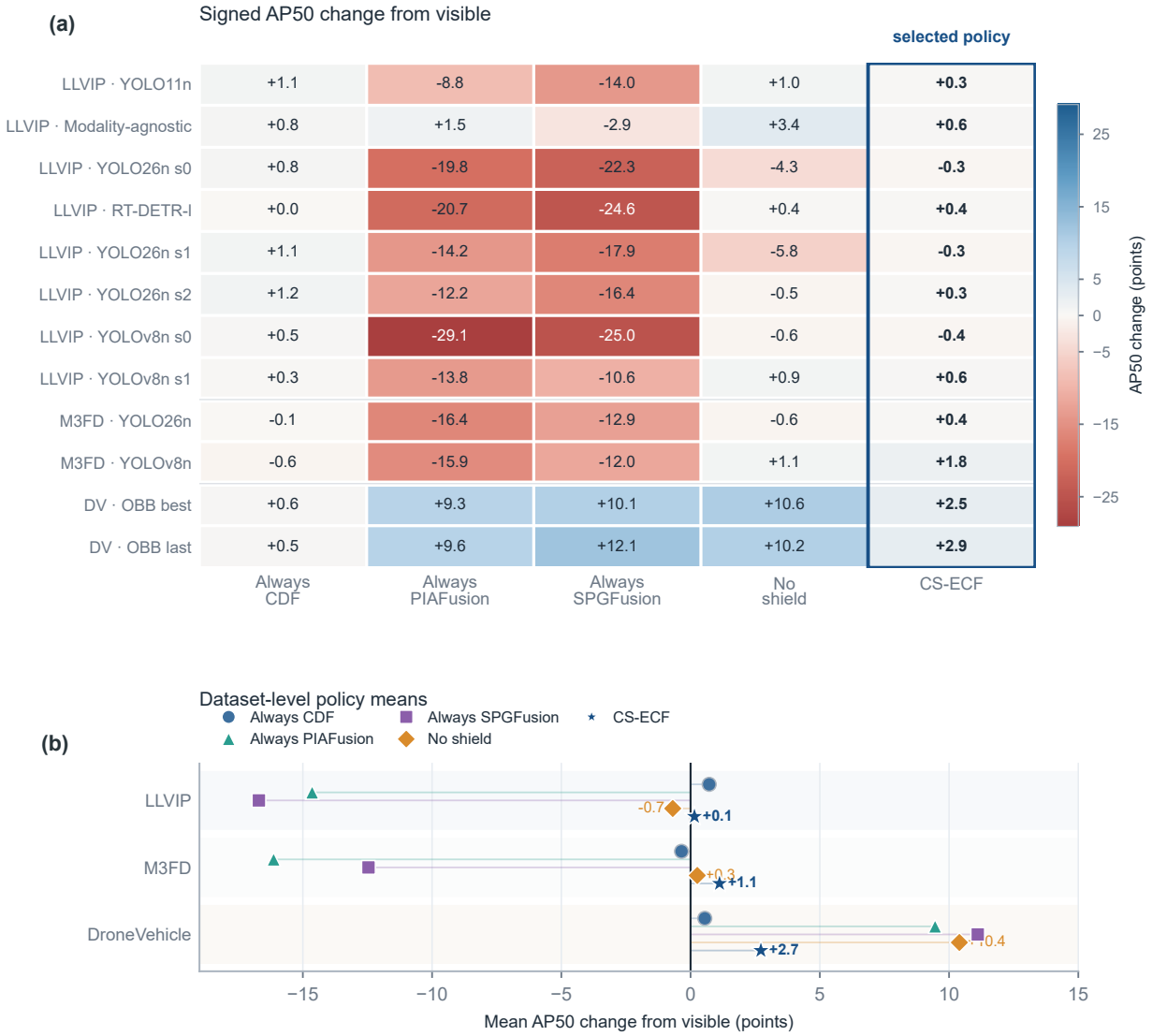


Figure 3: **Evaluator-dependent utility of infrared–visible fusion candidates.** (**a**), Signed AP50 change from the visible input for CDF, PIAFusion, SPGFusion, the no-shield selector, and CS-ECF across 12 detector checkpoints. Cells are point contrasts. (**b**), Dataset-level mean AP50 changes. The same candidate set produces opposite signs across datasets. Source data are provided in `Figure3_source.csv`.

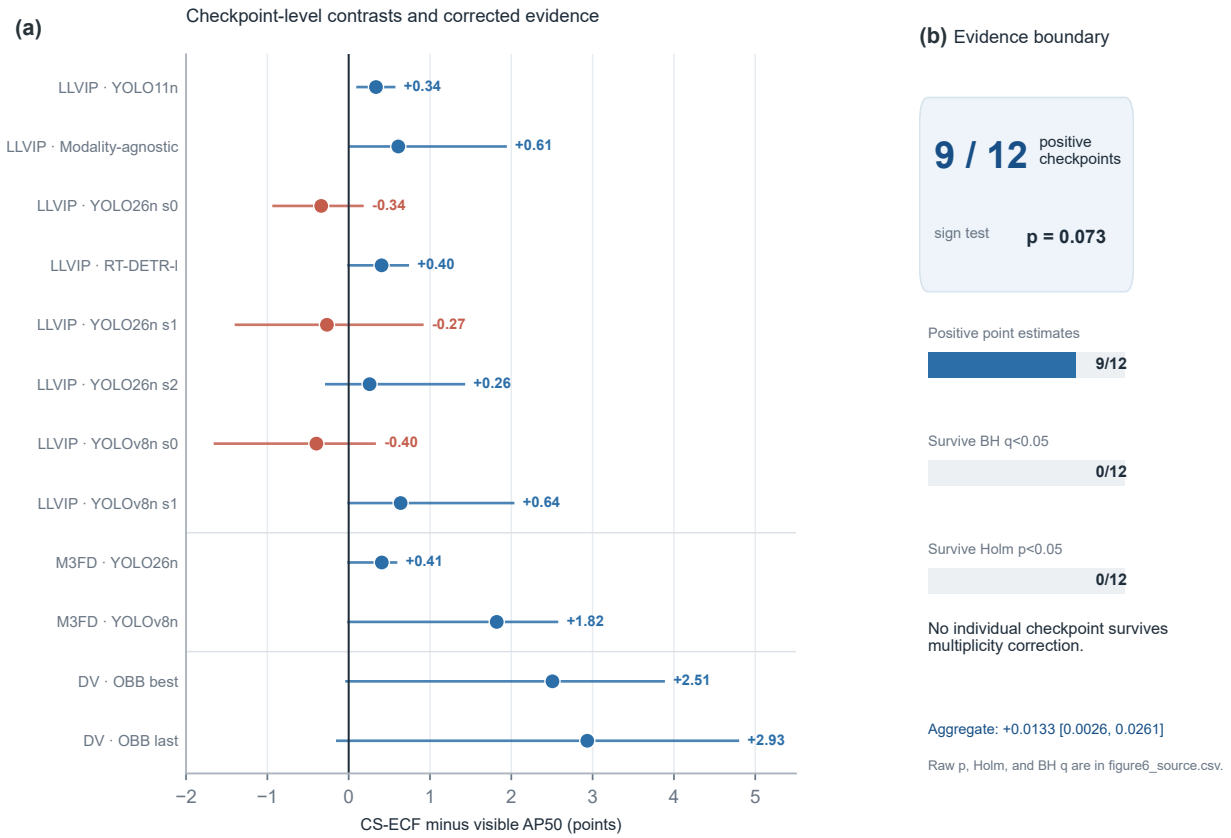


Figure 4: **Checkpoint-level evidence and multiplicity correction.** (**a**), CS-ECF-minus-visible AP50 changes and scene/frame-group 95% intervals for the 12 confirmatory checkpoints. Horizontal rules separate datasets. (**b**), Nine point estimates are positive, but the exact sign test gives $p = 0.073$, and no checkpoint survives Benjamini–Hochberg or Holm correction at 0.05. The dataset-balanced estimate is +0.0133 with interval $[0.0026, 0.0261]$. Raw p-values, Holm-adjusted p-values, and BH q-values are supplied in `Figure4_source.csv`.

Table 4: Confirmatory checkpoint summary and policy comparison. Mean is the unweighted average across the 12 confirmatory checkpoints. Dataset-balanced averages within each dataset and then gives LLVIP, M3FD, and DroneVehicle equal weight. Worst is the lowest checkpoint difference relative to visible.

Policy	Positive	Mean	Dataset-balanced	Worst
Always CDF	10/12	+0.0052	+0.0031	-0.0059
Always PIAFusion	3/12	-0.1087	-0.0710	-0.2910
Always SPGFusion	2/12	-0.1136	-0.0602	-0.2501
No-shield selector	7/12	+0.0132	+0.0332	-0.0584
CS-ECF	9/12	+0.0074	+0.0133	-0.0040

Table 5: Paired AP50 difference between CS-ECF and Always CDF. Dataset-specific intervals resample scene or frame groups. The combined interval also resamples training runs within each dataset and then weights datasets equally.

Scope	CS-ECF − Always CDF	95% group-aware interval
LLVIP	−0.0057	$[-0.0189, +0.0156]$
M3FD	+0.0147	$[+0.0100, +0.0159]$
DroneVehicle	+0.0217	$[-0.0007, +0.0738]$
Dataset-balanced	+0.0102	$[-0.00057, +0.02980]$

4.2. Dataset-specific behaviour

On LLVIP, the mean gain was 0.0016 and five of eight checkpoints improved; the checkpoint and image intervals crossed zero. Both M3FD six-class checkpoints improved, with a mean gain of 0.0111. On the independent DroneVehicle test subset, the best and last OBB checkpoints improved by 0.0251 and 0.0293, and the combined group interval was [0.0002, 0.0406]. All three DroneVehicle groups had non-negative point estimates, although one gained only 0.0002. The two OBB checkpoints originated from one training run and are therefore not independent architectures.

4.3. Direct comparison with the strongest fixed baseline

Among the fixed-input policies, Always CDF performed best and therefore provides the strongest baseline for evaluator-conditioned arbitration. Across the 12 checkpoints, CS-ECF exceeded Always CDF in six cases and fell below it in six. The dataset-balanced difference was +0.0102 AP50, but its training-run-cluster-by-group interval, $[-0.00057, 0.02980]$, included zero. LLVIP favoured Always CDF, M3FD had a positive point contrast, and DroneVehicle was represented by a single training-run cluster. Table 5 therefore retains the dataset-specific contrasts, and Fig. 5 visualizes them without averaging over the dataset structure.

This comparison narrows the claim. CS-ECF did not show a statistically separable improvement over Always CDF. Its average advantage was concentrated in the mixed-class and vehicle-focused evaluations and was negative on LLVIP. The defensible contribution is risk-controlled, evaluator-conditioned arbitration and its observed trade-off, not dominance over the strongest fixed input.

4.4. Controls and headroom

To determine whether detector confidence adds information beyond image quality, we evaluated label-free controls and a label-using upper bound. The VIF selector chose the fused candidate with the highest mean reference VIF relative to the visible and infrared inputs, and the SSIM selector used the analogous mean structural similarity. VIF and SSIM selected only among the three fused candidates, whereas entropy selection considered all four inputs. All image statistics were computed on 256×256 grayscale inputs, which made the measures comparable across source resolutions and detector architectures. At each checkpoint, we generated 500 uniform random candidate allocations across images. The oracle used test labels to choose the candidate with the

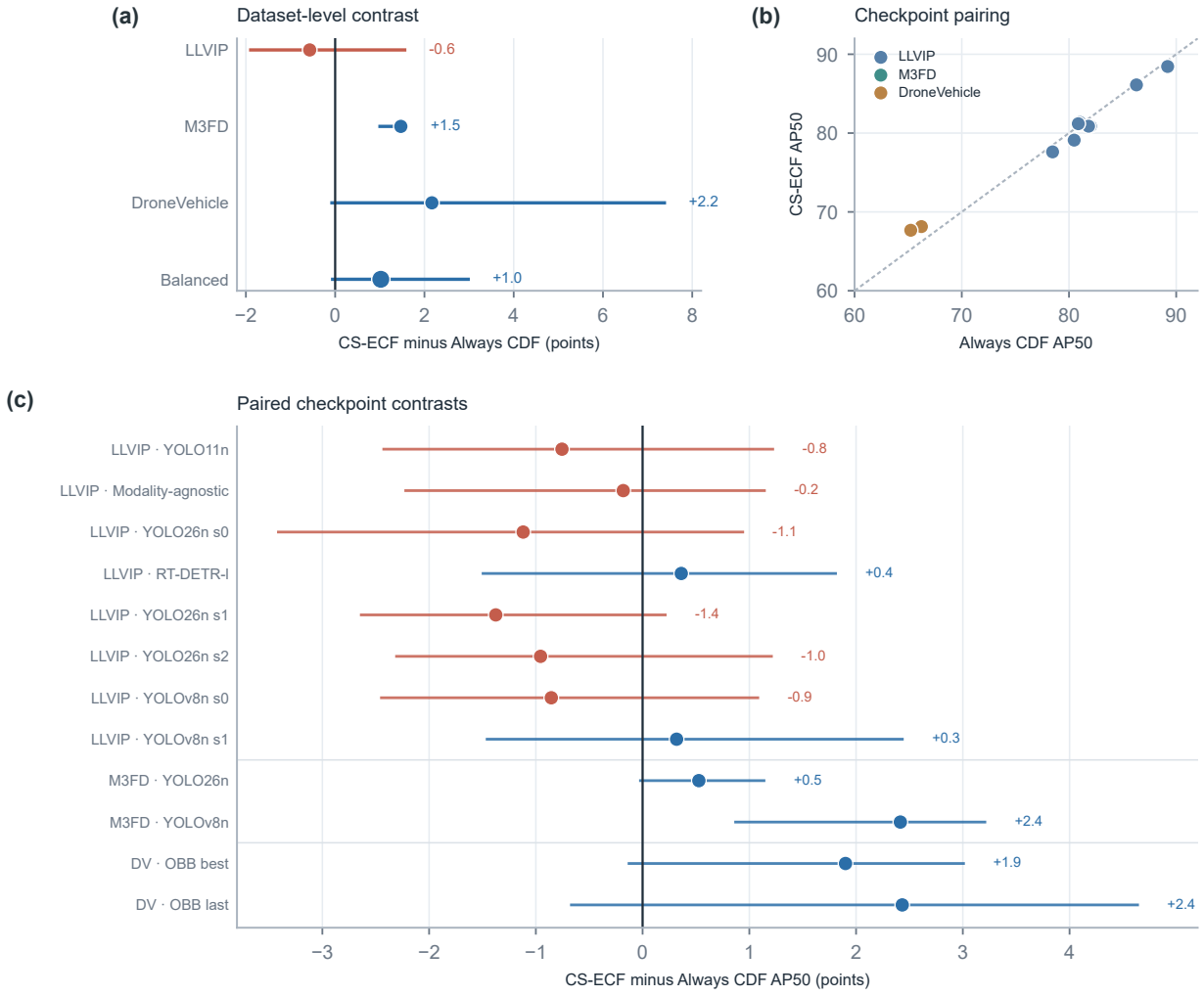


Figure 5: **CS-ECF versus the strongest fixed fusion baseline.** (**a**), Dataset-level and dataset-balanced CS-ECF-minus-Always CDF AP50 contrasts with 95% group-aware intervals. The dataset-balanced interval crosses zero. (**b**), Paired checkpoint AP50 positions relative to the equality line. (**c**), Paired checkpoint contrasts. The point estimates favour CS-ECF in six checkpoints and Always CDF in six; the comparison therefore does not establish dominance over Always CDF. Source data are provided in `Figure5_source.csv`.

Table 6: Additional policy controls. Dataset-balanced AP50 and worst-checkpoint changes are relative to visible. Positive counts refer to checkpoint-level point estimates. The random row is the mean of 500 uniform candidate allocations per checkpoint, and its worst entry is the lowest checkpoint mean. The oracle uses test labels and is an upper bound rather than a deployable selector.

Policy	Dataset-balanced AP50	Worst checkpoint	Positive
Always CDF	+0.0031	-0.0059	10/12
Uniform random	-0.0341	-0.1341	2/12
Maximum reference VIF	+0.0008	-0.0059	8/12
Maximum SSIM	-0.0946	-0.2855	3/12
Maximum entropy	-0.0599	-0.2432	2/12
CS-ECF	+0.0133	-0.0040	9/12
Per-image oracle	+0.0833	+0.0205	12/12

Table 7: External evaluator stress tests on the same 300 LLVIP test pairs. Values in the second column are visible / Always CDF / no-shield / CS-ECF AP50. The first two evaluators are COCO-pretrained without LLVIP fine-tuning; the third is LLVIP-fine-tuned and tests a negative portability boundary. Intervals resample LLVIP scene groups. NS, no-shield selector.

Evaluator	AP50 (V/CDF/NS/CS)	CS-ECF − visible	CS-ECF − CDF
YOLOv9t, zero-shot	.7698 / .7888 / .8216 / .7948	$+.0250$ $[+.0053, +.0523]$	$+.0060$ $[-.0277, +.0374]$
YOLOv10s, zero-shot	.8344 / .8617 / .8699 / .8557	$+.0213$ $[+.0041, +.0439]$	$-.0059$ $[-.0266, +.0180]$
YOLOv9t, fine-tuned	.6666 / .6598 / .6405 / .6558	$-.0108$ $[-.0342, +.0050]$	$-.0040$ $[-.0206, +.0108]$

highest per-image AP50 and is an upper bound only. Images without ground-truth boxes received an AP50 of zero for every candidate, so ties resolved to visible.

The maximum-reference-VIF selector achieved a dataset-balanced gain of 0.0008 and 8/12 positive checkpoints, well below the confidence-based policy. SSIM and entropy selection produced strongly negative balanced gains. CS-ECF captured about 40% of the headroom between the mean uniform-random policy and the per-image oracle on dataset-balanced AP50. Detector confidence therefore added value beyond image-quality heuristics, although substantial headroom remained (Fig. 6).

4.5. External evaluator stress tests

To test whether the result depended on the evaluator families used during policy development, we first applied the frozen selector to COCO-pretrained YOLOv9t and YOLOv10s checkpoints without LLVIP fine-tuning. We then tested the same selector with an LLVIP-fine-tuned YOLOv9t, whose lower test-set performance and larger train-to-test shift provide a stress case for the confidence score.

Both zero-shot evaluators improved in AP50 relative to visible, but neither showed a separable advantage over Always CDF. The LLVIP-fine-tuned YOLOv9t reversed the direction of the contrast: CS-ECF was 0.0108 AP50 below visible, with a scene-group interval crossing zero, and 0.0040 below Always CDF. The shield nevertheless improved the no-shield selector by 0.0153 AP50, with a group interval of $[+0.0072, +0.0267]$. It selected visible for 230 of 300 images and reduced the worst scene loss from 0.1208 to 0.0724, but did not eliminate that loss.

The shield's effect on the worst case was mixed across the zero-shot checks. On YOLOv10s, it reduced the worst scene loss from 0.0337 to 0.0038. On zero-shot YOLOv9t, the worst scene loss was 0.0167 with the shield and 0.0011 without it.

The LLVIP-fine-tuned checkpoint switched to fusion at a rate similar to the zero-shot evaluators, so switching frequency did not explain the sign reversal. Its calibration error was about five times larger than those of the zero-shot models (0.048 versus 0.009–0.010). Among the 70 images on which it selected fusion, 41.4% degraded and only 25.7% improved in per-image AP50. The corresponding degradation fractions for the two zero-shot evaluators were 20.9% and 9.1%. The fine-tuned model also produced 2 positive and 5 negative scene groups. Benjamini–Hochberg

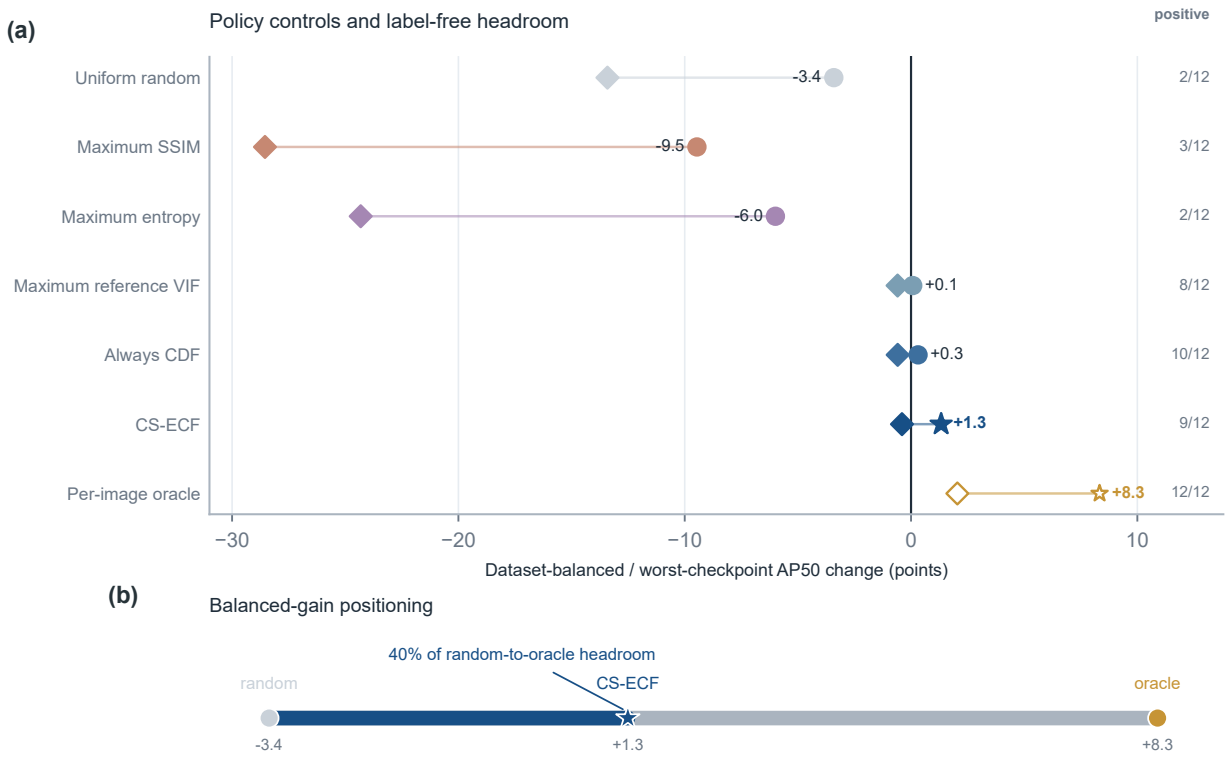


Figure 6: **Label-free controls and oracle headroom.** (**a**), Dataset-balanced and worst-checkpoint AP50 changes for uniform random selection, three image-quality selectors, Always CDF, CS-ECF, and a per-image oracle. The count at the right gives positive checkpoints out of 12. (**b**), Position of CS-ECF between the mean uniform-random policy and the label-using oracle. The oracle is an optimistic upper bound, not a deployable selector. VIF and SSIM selected only among fused candidates; entropy selected among all four inputs. Source data are provided in `Figure6_source.csv`.

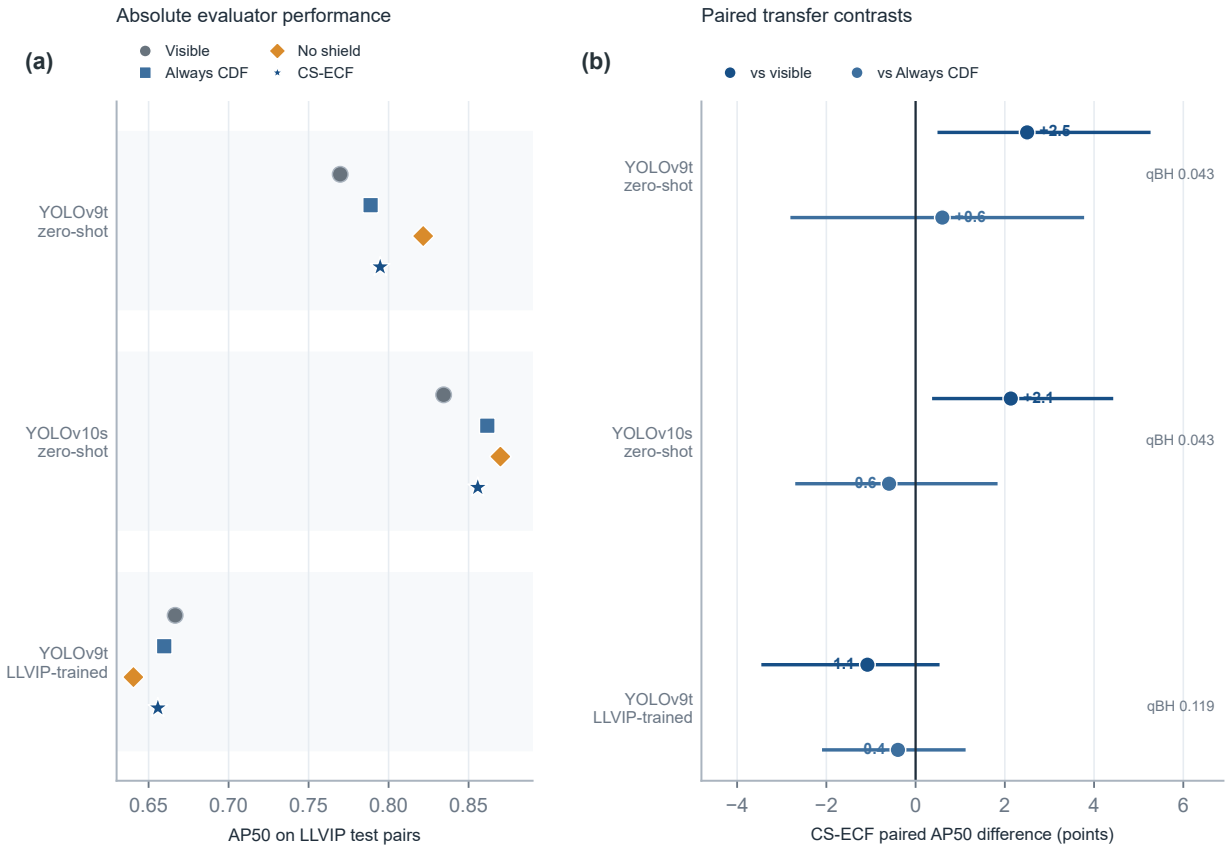


Figure 7: **External-evaluator portability stress tests.** (**a**), Absolute AP50 for visible, Always CDF, the no-shield selector, and CS-ECF under two zero-shot COCO evaluators and one LLVIP-fine-tuned YOLOv9t. (**b**), Paired CS-ECF contrasts against visible and Always CDF with scene-group 95% intervals. This externally defined evaluator set is analysed with Benjamini–Hochberg correction. The fine-tuned evaluator reverses the visible-relative sign. Source data are provided in `Figure7_source.csv`.

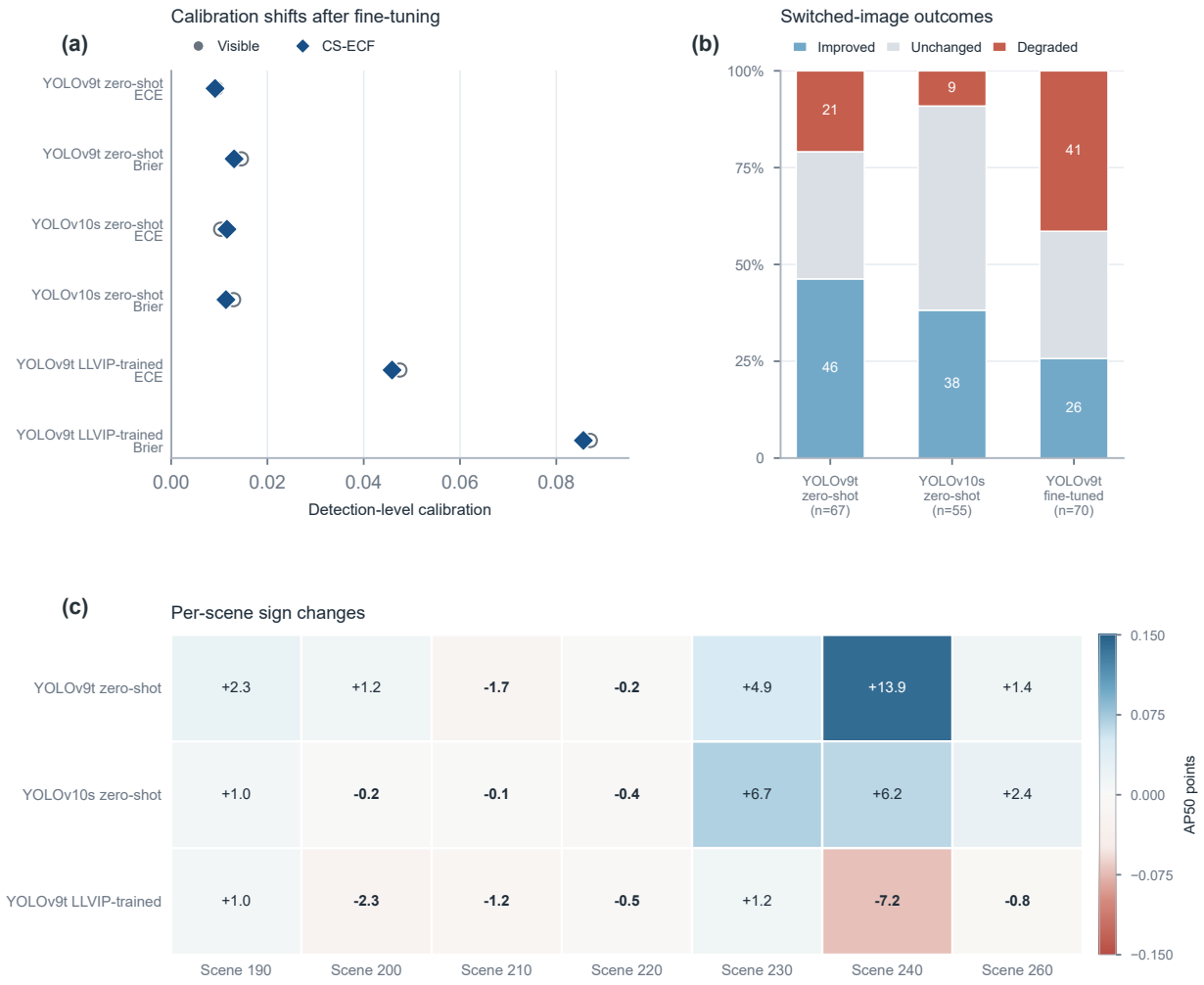


Figure 8: **Calibration and switched-image failure boundary.** (**a**), Detection-level ECE and Brier scores for visible and CS-ECF. (**b**), Improved, unchanged, and degraded per-image AP50 among images assigned to a fused candidate. (**c**), Scene-level CS-ECF-minus-visible AP50 changes. The LLVIP-fine-tuned evaluator has substantially higher calibration error, more degraded switches, and two positive versus five negative scene groups. Source data are provided in `Figure8_source.csv`.

Table 8: Diagnostics for the external-evaluator boundary. Decision fractions are visible/fused proportions over 300 images. ECE is the detection-level expected calibration error with ten fixed confidence bins. Switched outcomes are improved/unchanged/degraded per-image AP50 among images assigned to a fused candidate. Scene counts indicate positive/negative CS-ECF-minus-visible groups.

Evaluator	Visible/fused	ECE (visible/CS)	Switched I/U/D	Scenes +/−
YOLOv9t, zero-shot	77.7%/22.3%	0.009/0.009	46.3%/32.8%/20.9%	5/2
YOLOv10s, zero-shot	81.7%/18.3%	0.010/0.012	38.2%/52.7%/9.1%	4/3
YOLOv9t, fine-tuned	76.7%/23.3%	0.048/0.046	25.7%/32.9%/41.4%	2/5

correction across the three external contrasts gave $q = 0.043$, 0.043, and 0.119 for YOLOv9t zero-shot, YOLOv10s zero-shot, and YOLOv9t LLVIP-fine-tuned, respectively. This pattern is consistent with a calibration-related boundary, but the current design cannot establish causality (Figs. 7 and 8).

4.6. The shield is a risk-control mechanism

Without the shield, the selector frequently chose fused candidates on LLVIP and DroneVehicle and sometimes produced large confidence gains that did not improve AP. The 1.5 margin reduced the number of switches and lowered the magnitude of the worst AP50 loss from 0.0584 to 0.0040. On DroneVehicle, however, the shield also reduced the mean gain from 0.1040 to 0.0272. This trade-off captures the central risk-control result of the study (Fig. 9).

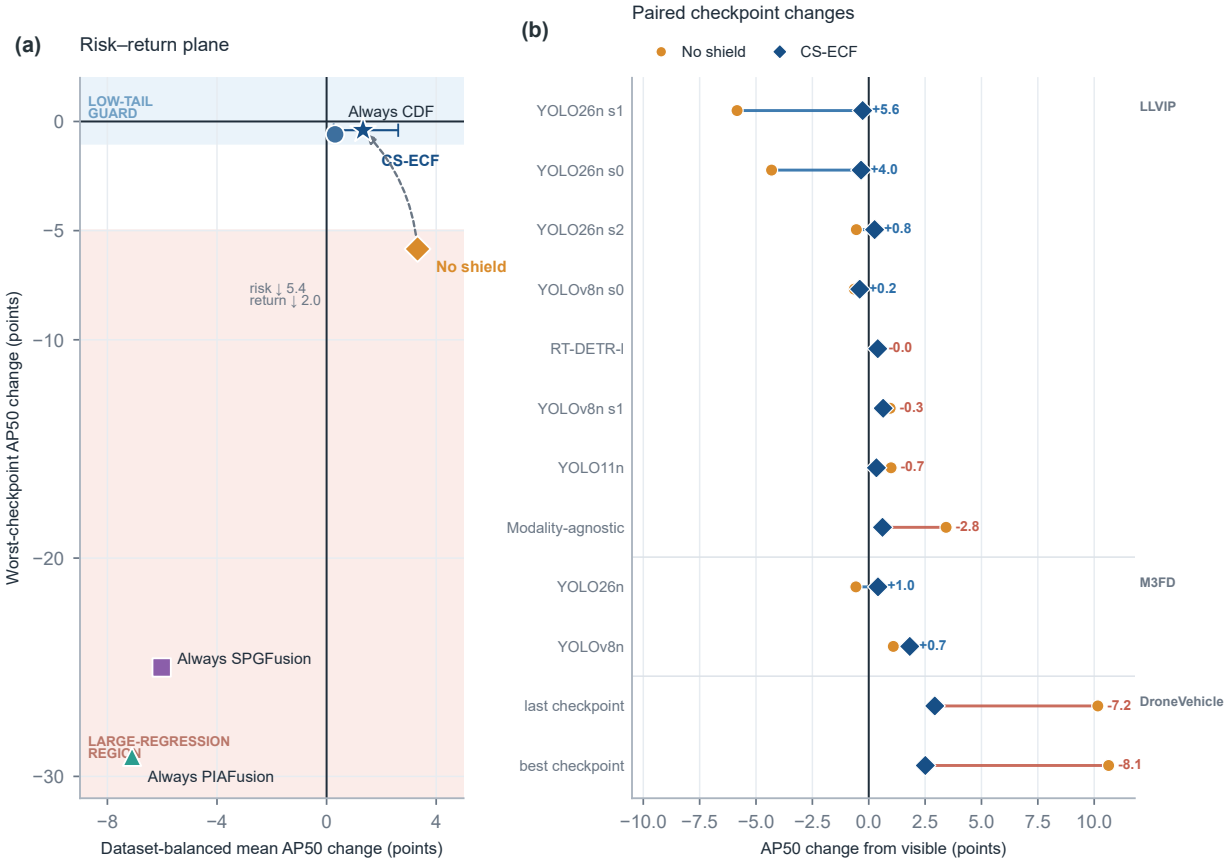


Figure 9: **Risk–return trade-off of fusion-input policies.** (**a**), Dataset-balanced mean AP50 change versus worst-checkpoint AP50 change. The CS-ECF horizontal error bar is the training-run-by-group hierarchical 95% interval. (**b**), Paired checkpoint changes under the no-shield selector and CS-ECF. The shield sacrifices part of the DroneVehicle mean gain to reduce the worst LLVIP loss. Source data are provided in `Figure9_source.csv`.

4.7. Fixed baselines and class concentration

At fixed inputs, Always CDF had positive AP50 changes at 10 of 12 checkpoints and a dataset-balanced gain of 0.0031, but remained negative on both M3FD detectors. Always PIAFusion and Always SPGFusion had negative dataset-balanced gains because their strong DroneVehicle performance did not transfer to the other datasets. In the M3FD six-class evaluation, the CS-ECF gain was concentrated in the people class: YOLO26n improved by 0.0214 and YOLOv8n improved by 0.1090. The other five classes changed by at most 0.0030 in absolute AP50 (Fig. 10). The people class accounts for only 166 of 2,022 test annotations (8.2%), but AP50 weights classes equally. It contributed 87.7% and 99.6% of the positive class-mean gain for the two detectors, respectively. Image-bootstrap intervals were $[-0.0070, +0.0498]$ for YOLO26n and $[+0.0440, +0.1913]$ for YOLOv8n; only the latter excluded zero. With only two test frame groups, these class intervals are descriptive rather than stable group-level inference. The evidence therefore supports a people-specific effect, not uniform six-class improvement.

4.8. Thermal feature association and mask ablation

On LLVIP, infrared top-hat contrast showed a weak positive association with fused-input selection (Spearman $\rho = 0.151$, $p = 0.009$), whereas visible luminance showed a weak negative association ($\rho = -0.104$, $p = 0.071$). On M3FD, thermal contrast showed a moderate positive association ($\rho = 0.320$, $p < 10^{-5}$). On DroneVehicle, both thermal contrast ($\rho = -0.403$) and visible luminance ($\rho = -0.441$) were negatively associated with selection. A high-thermal/low-visible median split did not produce a systematically larger AP gain than the remaining images.

A fixed-input ablation showed a small but consistent thermal-mask effect on DroneVehicle. The CDF gain was 0.0060 with the thermal-validity mask and 0.0057 with global detail injection for the best checkpoint. The corresponding values were 0.0050 and 0.0038 for the last checkpoint. On average, the thermal mask added 0.0007 AP50 relative to global injection.

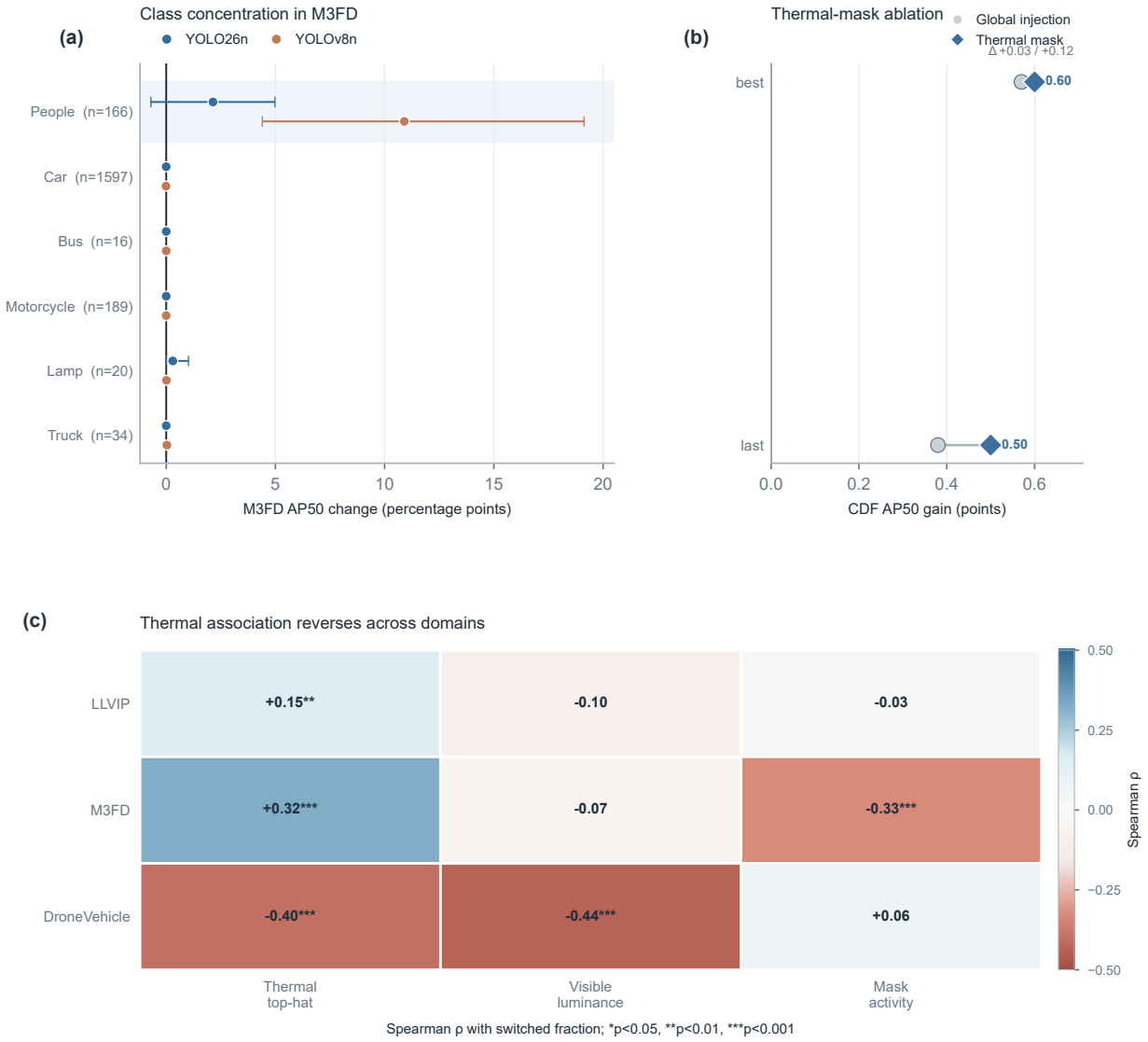


Figure 10: **Thermal constraints and class-specific effects.** (**a**), M3FD AP50 changes for six classes with image-bootstrap 95% intervals and annotation counts. The gain is concentrated in people. (**b**), Fixed-input DroneVehicle CDF gain with the thermal-validity mask or global detail injection. (**c**), Spearman association between the fused-selection fraction and thermal top-hat contrast, visible luminance, or CDF-mask activity; asterisks denote p-values below 0.05, 0.01, and 0.001, respectively. The association reverses on DroneVehicle, which does not support a universal thermal switching rule. Source data are provided in `Figure10_source.csv`.

4.9. Registration sensitivity

We estimated translation residuals for all 300 LLVIP test pairs using ECC on normalized Sobel edges. The median residual was 2.53 pixels. The median annotated person box was 109 pixels wide and 246 pixels high, so the median residual corresponded to 2.3% of the median box width and 1.0% of its height. On the 75 best-aligned pairs (residual at most 1.61 pixels), the mean CS-ECF gain across the eight LLVIP checkpoints was 0.0007, with five positive checkpoints. On the 75 worst-aligned pairs, the mean gain was 0.0020, with six positive checkpoints. Most checkpoint-specific intervals still crossed zero, so these subsets do not identify a distinct registration regime. They do show that the observed gain was not confined to the best-aligned quartile. The same residual audit was not available for M3FD or DroneVehicle.

5. Discussion

The central finding is that fusion utility depends on the evaluator and is not an intrinsic property of a fused image. One candidate can improve a detector's confidence ordering while

harming another's, and ignoring this dependence can lead to inputs that look confident but reduce AP. CS-ECF addresses this problem by using the target detector as an arbiter and requiring a large visible-relative margin before switching.

The shield's value is clearest in the worst case. On DroneVehicle, the no-shield selector achieved a higher mean gain where PIAFusion and SPGFusion were strong, but it also caused severe LLVIP losses. The 1.5 margin favours visible fallback under ambiguous evidence. This behaviour is consistent with a deployment setting in which a rare large regression is more costly than a missed moderate gain.

The comparison with Always CDF further narrows the use case. When CDF is stable for a fixed evaluator family, the simpler fixed input remains preferable. CS-ECF is more relevant when the evaluator or data domain can change which fusion family performs best, as observed across pedestrian, mixed-class, and vehicle-focused evaluation. Its modest balanced advantage should therefore not be read as evidence that it can replace CDF everywhere, because the interval included zero.

The multiplicity analysis further limits the strength of the evidence. Although the dataset-balanced interval excludes zero, no individual checkpoint contrast survived Holm correction, and the 9/12 sign test was not significant. These results support a small average effect under a prespecified composite estimand, not checkpoint-specific improvement.

The external evaluators provide a more heterogeneous picture. Both zero-shot checkpoints support a visible-relative gain but not superiority over Always CDF. The LLVIP-fine-tuned YOLOv9t reverses the visible-relative contrast, while shielding still improves on the no-shield selector. This checkpoint also had higher calibration error and a higher degradation rate among switched images. These observations are consistent with a calibration-related failure, but its lower test performance and larger train-to-test shift provide alternative explanations. The current design cannot separate these explanations.

Three experimental boundaries further constrain the interpretation. First, the strongest M3FD effect is class-specific: people detection improves substantially while the other classes remain nearly unchanged. Second, the DroneVehicle confirmation uses two checkpoints from one training run, so the checkpoints are correlated. Third, a target detector not included in confirmatory testing can reverse the sign of the visible-relative contrast, as observed for the LLVIP-fine-tuned YOLOv9t. The policy should therefore be validated for each deployed evaluator rather than assumed to be portable.

A further boundary concerns the LLVIP training subset. It was selected lexicographically rather than by scene-stratified sampling, so its scene composition may differ from that of the full official training partition. The test subset was sampled randomly from all 3,463 non-empty official test pairs.

The thermal analysis adds another limitation. Thermal contrast may increase the likelihood of selecting fused evidence on LLVIP and M3FD, but the association reverses on DroneVehicle. Sensor-level thermal evidence and detector confidence therefore capture different information, and neither is a universal utility score. The appropriate claim is detector-conditioned thermal fusion utility, not thermal dominance.

Compute cost is also substantial. SPGFusion dominates serial latency, and four detector passes add further cost. CS-ECF is therefore suited to settings in which the target detector is already run and a large input-selection regression is costly. It is not a low-latency replacement for a single fusion operator. Candidate pruning and early termination remain untested; their latency-accuracy trade-off requires a separate deployment study (Fig. 11).

A practical decision sequence follows from these boundaries:

1. use Always CDF when the detector is fixed and validation supports stable CDF performance;
2. consider CS-ECF when the evaluator or domain may change and the target model can be calibrated and revalidated on held-out groups;
3. use the shield when a large regression is more costly than a missed moderate gain;

4. avoid the four-candidate search for latency-sensitive deployment unless pruning or early termination is evaluated separately;
5. revalidate the threshold after fine-tuning materially changes confidence calibration.



Figure 11: **Runtime cost and deployment boundaries.** (**a**), Serial latency of candidate generation and four detector passes at the stated hardware and input sizes; the estimate excludes model swapping, data loading, and final post-processing. (**b**), Per-stage latency and extra peak memory, with marker area scaled by parameter count. Per-stage memory values are not additive. (**c**), Decision matrix implied by the observed evidence: Always CDF for a fixed, calibration-stable setting; CS-ECF after held-out revalidation when the evaluator or domain changes; visible or a stronger shield when calibration is unstable; and no four-candidate search for latency-sensitive deployment unless pruning or early termination is measured. Source data are provided in `Figure11_source.csv`.

6. Conclusions

We showed that infrared–visible fusion utility depends on the evaluator and proposed CS-ECF, a confidence-shielded arbitration policy with a fixed visible-relative margin and explicit visible fallback. Across three datasets and 12 checkpoints, nine checkpoint point estimates were positive, but none survived Holm correction and the exact sign test was not significant ($p = 0.073$). The dataset-balanced AP50 gain was 0.0133 with a 95% interval of [0.0026, 0.0261], while the magnitude of the worst observed loss decreased from 0.0584 to 0.0040. Against Always CDF, the balanced difference was +0.0102, but its interval included zero, so dominance was not established. Two zero-shot evaluators improved AP50 over visible by 0.021–0.025 without separating from Always CDF. The LLVIP-fine-tuned YOLOv9t lost 0.0108 AP50 relative to visible, although the shield

improved the no-shield selector by 0.0153. These results support bounded, evaluator-dependent risk control for pedestrian-centric and mixed infrared–visible detection. They do not support uniform six-class improvement, universal portability across evaluators, independent replication of the DroneVehicle result from one training run, or dominance over Always CDF. Deployment also requires consideration of SPGFusion latency and the cost of four detector passes.

CRediT authorship contribution statement

Jianyong Zhao: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Zigao Wu:** Supervision, Writing – review & editing. **Mengfei Ma:** Investigation, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Data availability

The underlying datasets are publicly available. LLVIP is distributed at `https://github.com/bupt-ai-cz/LLVIP`. M3FD is distributed with the TarDAL benchmark at `https://github.com/JinyuanLiu-CV/TarDAL`. DroneVehicle is distributed through `https://github.com/VisDrone/DroneVehicle` and was obtained from the public archive `https://huggingface.co/datasets/McCheng/DroneVehicle`. Each dataset is used under the licence stated by its distributor. The public analysis repository described below contains the fixed DroneVehicle test-group manifest and source-archive metadata. It also includes CS-ECF per-image score files, source-data tables, and scripts that regenerate the reported intervals.

Code availability

The code for reproducing the experiments is available at `https://github.com/lessThanbetter/ssgf` under an MIT licence. The version-specific archival release is available at `https://doi.org/10.5281/zenodo.23728385`. The repository and archive contain the fusion and arbitration code, detector-evaluation scripts, fixed split and test manifests, and per-image score files. They also include result tables, figure source data, and numerical checks used to generate the reported intervals.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex to support language editing, data-analysis code development, figure preparation, and the organization of numerical results. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The detector models, numerical results, and reported conclusions were verified by the authors.

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